

Problem A - Apartment Tycoon

In the bustling metropolis of Econopolis, where the skyline is ever-changing and the economy thrives on savvy investments, a young entrepreneur embarks on a journey to become a real estate mogul. Gifted with a single apartment by a mysterious benefactor, this aspiring tycoon sees an opportunity to expand their empire. The strategy is simple yet ambitious: use the rental income from the apartment to acquire more, each becoming a new source of revenue. The goal? To amass a portfolio of C apartments, transforming the initial gift into a sprawling empire. The journey from a single property to a real estate dynasty will require cunning, patience and a keen eye for opportunity.

Given that you start with one apartment generating B dollars per month and can purchase additional apartments for A dollars each, which also start generating B dollars per month once acquired, your task is to calculate how many months it will take to own C apartments. Each month, you can use the income generated to buy more apartments, accelerating your journey toward becoming an apartment tycoon.

Input

The input consists of a single line containing three integers, A , B , and C ($1 \leq A, B, C \leq 1000$), representing the cost of an apartment, the monthly income from each apartment, and the target number of apartments, respectively.

Output

Print a single integer, the number of months required to own C apartments.

Sample input 1 5 1 2	Sample output 1 5
Sample input 2 10 5 3	Sample output 2 3